Stage 1 Hard Drill — Question + Solution

Chapter 1.3 — Univariate Discrete & Continuous Distributions

Full-chapter coverage, 10 problems, each with its complete derivation attached.

Q1 — Binomial: Mode-Tie Trap

Question. A market-making bot sends $n = 24$ independent quotes, each filled with probability p. For exactly which value(s) of $p = k/25$ (integer k) does the fill-count distribution have **two** modes instead of one, and what are they when $p = 0.6$?

Solution. Two modes $(n+1)p - 1$, $(n+1)p$ occur exactly when $(n+1)p \in \mathbb{Z}$. Here $(n+1)p = 25 \cdot \frac{k}{25} = k$ — an integer *automatically*, for every $k = 1, \ldots, 24$.

Every $p = k/25$ with $k = 1, \ldots, 24$ gives a tie (forced by $n+1 = 25$, not a coincidence). At $p = 0.6 = 15/25$: modes are $\boxed{14 \text{ and } 15}$.

Trap: the phrasing implies a special value; the lesson is that any p that's a multiple of $\frac{1}{n+1}$ guarantees a tie.

Q2 — Poisson Approximation: Judgment, Not Computation

Question. Desk A: $n = 200, p = 0.02$. Desk B: $n = 20, p = 0.2$. Both have $np = 4$. Without computing either distribution exactly, explain which Pois(4) approximation is trustworthy and which is dangerous — then state the one-line criterion that decides it.

Solution. Compare exact variance $np(1-p)$ to the Poisson surrogate $\lambda = np = 4$: Desk A: $200(0.02)(0.98) = 3.92$ vs. 4 (2% error). Desk B: $20(0.2)(0.8) = 3.2$ vs. 4 (20% error).

Criterion: validity needs p itself small ($n \geq 20$, $p \leq 0.05$), *not* just a matched $\lambda = np$ — as p grows, $(1-p)$ drifts from 1 and the approximation's core assumption breaks.

Answer: Desk A trustworthy; Desk B dangerous despite identical λ.

Q3 — Geometric: Memoryless + Conditioning Trap

Question. A strategy fires once per day, independently, with probability $p = 0.1$. Given it has **not** fired in the first 5 days, find $\mathbb{E}$[additional days until it fires]. Harder: given it fires for the **first time on an even-numbered day**, find $\mathbb{E}$[that day].

Solution (Part 1, memoryless). $\mathbb{E}[\text{additional} \mid \text{not fired in 5}] = \mathbb{E}[T] = \frac{1}{p} = \boxed{10}$ days.

Solution (Part 2, parity — NOT memoryless). Let $q = 0.9$. With $\mathbb{P}(T = 2m) = q^{2m-1}p$:

$$\mathbb{P}(T \text{ even}) = \frac{qp}{1-q^2} = \frac{q}{1+q}, \qquad \mathbb{E}[T \cdot \mathbb{1}\{T \text{ even}\}] = \frac{2pq}{(1-q^2)^2}.$$

$$\mathbb{E}[T \mid T \text{ even}] = \frac{2}{p(1+q)} = \frac{2}{p(2-p)} \approx \boxed{10.53} \text{ days at } p = 0.1.$$

Why not memoryless: parity conditioning constrains every preceding odd trial, unlike a fixed elapsed-time condition — the closeness of 10 vs. 10.53 is the trap.

Q4 — Negative Binomial: Decomposition Speed Trick

Question. A quant needs 5 independent model validations, each succeeding with probability $p = 0.4$. Using the sum-of-Geometrics decomposition, find $\mathrm{Var}(N)$, the total attempts needed.

Solution. $N = G_1 + \cdots + G_5$, i.i.d. $G_i \sim \mathrm{Geom}(0.4)$.

$$\mathrm{Var}(G_i) = \frac{1-p}{p^2} = \frac{0.6}{0.16} = 3.75 \;\Rightarrow\; \mathrm{Var}(N) = 5(3.75) = \boxed{18.75}.$$

Why "free": the G_i are i.i.d. and independent because each gap depends only on fresh trials after the previous success. Under dependence, cross-covariance terms would reappear and destroy the simple sum.

Q5 — Poisson: Sum + Thinning Combined

Question. Orders arrive as Pois(12)/hr. Each is independently small (0.25), medium (0.5), or large (0.25). Find $\mathbb{P}(\geq 1$ large AND exactly 2 small in an hour).

Solution. Thinning: $\lambda_{\text{small}} = \lambda_{\text{large}} = 12(0.25) = 3$, independent Poissons (by the thinning theorem).

$$\mathbb{P}(\geq 1 \text{ large}) = 1 - e^{-3} \approx 0.9502, \qquad \mathbb{P}(\text{exactly 2 small}) = 4.5e^{-3} \approx 0.2240.$$

$$\mathbb{P}(\text{both}) = 0.9502 \times 0.2240 \approx \boxed{0.2129}.$$

Justification: thinning a Poisson process by i.i.d. labels yields independent sub-counts — provable via the joint PGF factoring cleanly, regardless of any apparent correlation through the shared stream.

Q6 — Uniform: Nonlinear Transformation Trap

Question. $X \sim \mathrm{Unif}(0,1)$. Let $Y = \min(X, 1-X)$. Find the PDF of Y and $\mathbb{E}[Y]$.

Solution. For $y \in [0, \frac{1}{2}]$: $\{Y \leq y\} = \{X \leq y\} \cup \{X \geq 1-y\}$, disjoint, so

$$F_Y(y) = 2y \;\Rightarrow\; f_Y(y) = 2, \; y \in [0, \tfrac{1}{2}].$$

This is exactly $\mathrm{Unif}(0, \frac{1}{2})$ — $\mathbb{E}[Y] = \boxed{1/4}$.

Honest note: the shape-guess was actually right (the fold has unit slope, preserving uniformity). The real trap is forgetting the density doubles to 2, since two disjoint tails feed each y.

Q7 — Uniform → Exponential: Inverse-Transform in Disguise

Question. $U \sim \mathrm{Unif}(0,1)$. Define $W = -\frac{1}{\lambda}\ln(1-U^2)$. Identify the distribution of W.

Solution. $W \leq w \iff U \leq \sqrt{1-e^{-\lambda w}}$, so

$$F_W(w) = \sqrt{1-e^{-\lambda w}}, \qquad f_W(w) = \frac{\lambda e^{-\lambda w}}{2\sqrt{1-e^{-\lambda w}}}, \; w \geq 0.$$

W is NOT Exp(λ). The inverse-transform fact $-\frac{1}{\lambda}\ln(1-V) \sim \mathrm{Exp}(\lambda)$ requires $V \sim \mathrm{Unif}(0,1)$. Here $V = U^2$ has CDF $\sqrt{v}$ (i.e. $\mathrm{Beta}(\frac{1}{2}, 1)$) — not uniform. *Lesson:* always verify the inner argument is Uniform before invoking the trick.

Q8 — Exponential: Racing with a Conditioning Twist

Question. Three market makers' clocks fire as independent $\mathrm{Exp}(\lambda_1), \mathrm{Exp}(\lambda_2), \mathrm{Exp}(\lambda_3)$. Given MM 1 updates **last**, find $\mathbb{E}$[time until MM 1 updates].

Solution. Recursive-race decomposition. Let $\lambda = \lambda_1 + \lambda_2 + \lambda_3$, $a = \frac{\lambda_2}{\lambda} \cdot \frac{\lambda_3}{\lambda_1+\lambda_3}$, $b = \frac{\lambda_3}{\lambda} \cdot \frac{\lambda_2}{\lambda_1+\lambda_2}$, and define the per-branch stage-time sums

$$c_1 = \frac{1}{\lambda} + \frac{1}{\lambda_1+\lambda_3} + \frac{1}{\lambda_1}, \qquad c_2 = \frac{1}{\lambda} + \frac{1}{\lambda_1+\lambda_2} + \frac{1}{\lambda_1}.$$

Then

$$\mathbb{P}(T_1 = \max) = a + b, \qquad \mathbb{E}[T_1 \mid \max] = \frac{a\,c_1 + b\,c_2}{a+b}.$$

Check: at $\lambda_1 = \lambda_2 = \lambda_3 = \lambda$, this gives $\mathbb{E}[T_1 \mid \max] = \frac{11}{6\lambda} = H_3/\lambda$ — matches Q9. *Trap:* $\mathbb{E}[T_1 \mid \max] \neq 1/\lambda_1$; conditioning on finishing last always inflates the expectation.

Q9 — Exponential: Harmonic Numbers, Extended

Question. Redo $\mathbb{E}[\max]$ of n i.i.d. $\mathrm{Exp}(\lambda)$ via memoryless-recursion, then find $\mathrm{Var}(\max)$ the same way.

Solution. $\max = \sum_{j=1}^{n} W_j$, $W_j \sim \mathrm{Exp}(j\lambda)$ mutually independent.

$$\mathbb{E}[\max] = \sum_{j=1}^{n} \frac{1}{j\lambda} = \frac{H_n}{\lambda}, \qquad \mathrm{Var}(\max) = \sum_{j=1}^{n} \frac{1}{(j\lambda)^2} = \boxed{\frac{H_n^{(2)}}{\lambda^2}}, \; H_n^{(2)} = \sum_{j=1}^{n} \frac{1}{j^2}$$

Independence of the W_j lets variances add exactly as the means did.

Q10 — Normal: Correlated Twist + Fat-Tail Judgment

Question. Strategy A$\sim \mathcal{N}(2\%, 5\%^2)$, B$\sim \mathcal{N}(1\%, 3\%^2)$, $\text{Corr}(A, B) = 0.4$. Portfolio: 60% A, 40% B. (a) Find σ_Y^2. (b) Does "$P(|Z| > 4)$ every $\sim$15,000 days" still apply?

Solution. $\text{Cov}(A, B) = 0.4(5)(3) = 6\,(\%^2)$.

$$\sigma_Y^2 = (0.6)^2(5)^2 + (0.4)^2(3)^2 + 2(0.6)(0.4)(6) = 9 + 1.44 + 2.88 = \boxed{13.32\,\%^2}$$

$$\sigma_Y \approx \boxed{3.65\%}$$

(vs. 3.23% independent — positive correlation reduces diversification).

(b) No: both the empirical rule and the "1-in-15,000-days" figure assume an independent, exactly-Gaussian model. Correlation already raises σ_Y, and real returns carry excess kurtosis — so the true frequency of a $\geq 4\sigma$ move is strictly higher on *both* counts.

Q6 and Q7 are a matched pair: one "trap" that resolves as expected, one that doesn't. Verifying rather than pattern-matching is the tested skill.